CIP-A107 SOC Laboratory

Evidence Manifest and Findings Matrix

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1. Chronological evidence manifest

Screenshots are grouped by time block here; one row per screenshot with its SHA-256 is in screenshots_manifest.csv and SHA256SUMS.txt. Text exports carry their own SHA-256. "No" in the last column means the item is not part of the package.

Time	Artefact	Filename / exhibits	SHA-256	Source	Tool	In package
2026-10-15 11:28 to 14:00	Early network faults: no IPv4 on ens33, bridged to NAT, DHCP lease	E01 to E04 (4 photos)	see screenshots_manifest.csv	phone photos of the VM console and VMware settings	VMware, VM console	Yes
2026-10-15 14:09:26 to 14:43:35	Installer run 1 and failure	E05 to E06	see screenshots_manifest.csv	photos	VM console	Yes
2026-10-15 to 2026-10-18 19:09	Resource check, swap, snapshot Wazuh_Clean_Baseline (19:07:29 UTC), VM settings 8.6 GB / 4 vCPU	E07 to E10	see screenshots_manifest.csv	photos	VMware, VM console	Yes
2026-10-18 19:19:22 to 19:20:09	Run 2: GPG key error; resources and disk growth; DHCP repair	E11 to E13	see screenshots_manifest.csv	photos	VM console	Yes
2026-10-18 19:29:08 to 20:03:37	Run 3 failure; netplan name fix; download errors	E14 to E16	see screenshots_manifest.csv	photos	VM console	Yes
2026-10-18 20:33 to 22:03:20	apt tuning; run 4 and its failure at Filebeat	E17 to E20	see screenshots_manifest.csv	photos	VM console	Yes
2026-10-18 about 22:05 to 22:30	Key check, pre-download of packages (185 MB), package copy	E21 to E25	see screenshots_manifest.csv	photos	VM console	Yes
2026-10-18 22:32:46 to 22:48:01	Run 5 start and successful summary (password redacted)	E26 to E27	see screenshots_manifest.csv	photos	VM console	Yes
2026-10-18 22:48 to 23:20	Service and port check; memory check; indexer timeout (23:01:57); restart	E28 to E31; T13_indexer_timeout_dropin.txt	see screenshots_manifest.csv	photos; rebuilt text	VM console	Yes

Time	Artefact	Filename / exhibits	SHA-256	Source	Tool	In package
2026-10-18 23:24 to 23:48	SSH timeout, ufw, VMnet8 repair, new VM address 192.168.119.128, connectivity tests, login page	E32 to E40; T01_windows_network_diagnostics_before_fix.txt	see screenshots_manifest.csv	photos; original text upload	PowerShell, VM console, browser	Yes
2026-10-19 15:25 (manager)	Agent registration on the manager	T02_manage_agents_transcript.txt; T03_agent_control_list.txt	a4ffaba5a4506a5fea9fcd74f2f7e6c200ff23f79f5e3df32a72edf36ebaa9db	rebuilt text (key removed)	manage_agents, agent_control	Yes
2026-10-19 15:26 to 15:55	Agent download, install, key and IP saved, connection, dashboard shows Active	E42 to E44; T04_windows_agent_install_and_verify.txt	2a1fbfa9311eee2c4d5ab50de0889e1ce01dc7be7ee36c0c38575f6de67813e9	screenshots; rebuilt text	PowerShell, Agent Manager, dashboard	Yes
2026-10-19 15:39 to 15:45	FIM folder, config backup, one-line edit, restart, real-time monitoring started	T05_fim_configuration_and_restart.txt; T06_compare_object_ossec_conf.txt	95a062dee756d78c51d89573412dcf1116db68de35a3e744c65d0a7cf6e2319a	rebuilt text	PowerShell	Yes
2026-10-19 15:50 to 15:52	FIM run 1 (create, modify, delete) and its alerts	T07_fim_run1_script_output.txt; E46	28872d54385cf06189b463ac5a6674fa194092e369d0266513a299ba6c98436e	original text upload; screenshot	PowerShell, dashboard	Yes
2026-10-19 15:58:10 to 15:59:40	FIM run 2 and its alerts	T08_fim_run2_script_output.txt; T09_Lab05_FIM_run2_log.txt; T10_alert_554_added.json; T11_alert_550_modified.json; E47	5f01f1463f998bc1b96df1636a254abd9206fe22d129884e0767978a1ff520eb	rebuilt text; original JSON uploads; screenshot	PowerShell, dashboard	Yes
2026-10-19 after 16:00 (time not shown)	Manager alerts.log rule check	T12_manager_alerts_log_rule_ids.txt	0ce7c3e782f429db28afb8bf40c6c3d396ee3f85fb9962a3020331df159580b3	rebuilt text	grep on the manager	Yes
2026-10-19 about 16:13	Dashboard overview (agents summary, 24-hour alert counts)	E41	61976d1eb315f478ae38c7263220181dbf9b7a44384db573284e8385c34764ce	screenshot	dashboard	Yes
not recorded	Installer wazuh-install.sh, apt packages, agent MSI	wazuh-install.sh, wazuh-*_4.12.0-1_amd64.deb, wazuh-agent-4.12.0-1.msi	not recorded in the VM before removal	official Wazuh download	curl, apt, BITS	No (third-party installers are not submitted)
not captured	Inventory data pages of the dashboard (host name, board serial, installed software)	2 screenshots	n/a	screenshots	dashboard	No (personal and hardware identifiers; not needed)

2. Findings matrix

Confidence: High = shown directly by a tool or screenshot; Medium = one source or partly inferred; Low = proposal or unconfirmed. Source artefacts are exhibit (E) and text (T) numbers.

ID	Type	Finding	Source artefact	Confidence	Limitation
F01	Deployment	Wazuh 4.12.0 all-in-one (manager, indexer, dashboard, Filebeat) installed on the Ubuntu 20.04 VM; installer reported "Installation finished" at 22:48:01 UTC on 18 Oct.	E27; install_run5 log lines	High	Run 5 was the fifth attempt; the administrator password line is redacted.
F02	Deployment	The installer option -i printed "Hardware checks ignored"; in this version -i skips the hardware check and is not an interactive mode as the manual states.	E11, E26	High	Read from log lines of runs 2 and 5.
F03	Deployment	After run 5 the manager, dashboard and Filebeat were active and the indexer had failed; ports 1514, 1515, 55000 and 443 were listening.	E28	High	Status shown at one moment only.
F04	Deployment	The indexer failed on boot with "start operation timed out" at 23:01:57 UTC; a manual start succeeded and 9200 listened on 127.0.0.1.	E30, E31	High	Root cause (slow start under load) is inferred, not measured.
F05	Deployment	A systemd drop-in with TimeoutStartSec=900 was created to prevent a repeat.	T13_indexer_timeout_dropin.txt	High	Not tested by a second reboot.
F06	Troubleshooting	Run 1 failed at 14:41:23 UTC (15 Oct) because the dashboard web application did not initialise; the VM had 4.3 GiB RAM and 2 vCPU; no out-of-memory kill was found.	E06, E07	Medium	Resource shortage is the likely cause, not proven; later runs also changed other things.
F07	Troubleshooting	Run 2 failed with "Cannot import Wazuh GPG key" because ens33 had no IPv4 address at that moment.	E11, E13	High	
F08	Troubleshooting	The netplan file named the interface enp0s3 while the real interface is ens33; it was corrected with sed and netplan apply.	E14, E15	High	That the wrong name caused every lost lease is inferred.
F09	Troubleshooting	Runs 3 and 4 failed on package downloads (connection refused, IPv6 unreachable) from packages.wazuh.com; run 4 spent 72 minutes on the indexer before failing at Filebeat (21:59:55 UTC).	E16, E19, E20	High	The cause is the internet path; the remote server was not examined.
F10	Troubleshooting	Packages were pre-downloaded with apt-get --download-only (185 MB in 9 min 40 s at 318 kB/s) and copied to ~/wazuh-debs; run 5 then completed in 15 minutes.	E24, E25, E26	High	The installer also used the local apt cache; which files it took is not shown.

ID	Type	Finding	Source artefact	Confidence	Limitation
F11	Network	Windows could not reach the VM (ssh, Test-NetConnection time out) because the host adapter VMware Network Adapter VMnet8 was "Not Present"; Windows was only on Wi-Fi.	E32; T01_windows_network_diagnostics_before_fix.txt	High	Why the adapter disappeared is not known.
F12	Network	After Virtual Network Editor > Restore Defaults, VMnet8 was Up at 192.168.119.1 and the VM leased 192.168.119.128 (earlier 192.168.152.131).	E35, E36	High	The new NAT subnet came from the restore, not from a planned design.
F13	Access	The dashboard login page loaded at https://192.168.119.128/app/login with a "Not secure" warning (self-signed certificate); ports 443 and 22 tested True from Windows.	E40, E38	High	Certificate chain not examined.
F14	Agent	Agent 001 "Windows-SOC-Lab" was created with manage_agents (IP any); a repeated Add attempt was refused as a duplicate name.	T02_manage_agents_transcript.txt	High	
F15	Agent	The agent connected to 192.168.119.128:1514/tcp at 08:34:16 local (15:34:16 UTC); dashboard and agent_control show it Active (Windows 11 Pro 10.0.26200, v4.12.0).	E43, E44; T03_agent_control_list.txt; T04_windows_agent_install_and_verify.txt	High	
F16	FIM	The only change made to ossec.conf was one <directories realtime="yes" check_all="yes" report_changes="yes"> line for SensitiveFiles (Compare-Object).	T06_compare_object_ossec_conf.txt; T05_fim_configuration_and_restart.txt	High	check_all and report_changes go beyond the manual (realtime only).
F17	FIM	After restart, ossec.log at 08:45:43 local shows the SensitiveFiles path monitored with hash_md5, hash_sha1, hash_sha256, report_changes and realtime.	T05_fim_configuration_and_restart.txt	High	
F18	FIM	Run 1: three alerts for secret_plan.txt at 08:50:33.922 (added, level 5), 08:51:19.230 (modified, level 7) and 08:52:04.227 (deleted, level 7) local.	E46; T07_fim_run1_script_output.txt	High	No local record of hashes for run 1 (log folder missing).
F19	FIM	Run 2: alerts at 15:58:10.341, 15:58:55.443 and 15:59:40.419 UTC, 0.34 to 0.44 s after the script actions at 15:58:10, 15:58:55 and 15:59:40.	E47; T08_fim_run2_script_output.txt; T10_alert_554_added.json; T11_alert_550_modified.json	High	Script times have 1 s resolution.
F20	FIM	Rule IDs and levels are 554 (5), 550 (7) and 553 (7); the manager alerts.log lists the same two sequences.	T12_manager_alerts_log_rule_ids.txt; T10_alert_554_added.json; T11_alert_550_modified.json	High	
F21	FIM	Alert values match the local record: after create size 31 and SHA-256 482BC551…6C1A5D; after modify size 82 and SHA-256 EADB58EB…13085C; sha256_before of the modified alert equals the create hash.	T08_fim_run2_script_output.txt; T10_alert_554_added.json; T11_alert_550_modified.json	High	Only SHA-256 and size were compared with the local record; MD5 and SHA-1 come from the alert only.
F22	FIM	The modified alert reports changed attributes size, mtime, md5, sha1, sha256, and syscheck.diff shows the added line.	T11_alert_550_modified.json	High	

ID	Type	Finding	Source artefact	Confidence	Limitation
F23	FIM	The alert maps to MITRE T1565.001 (Stored Data Manipulation, tactic Impact) and to PCI DSS 11.5, GDPR II_5.1.f, HIPAA 164.312.c.1/c.2, NIST 800-53 SI.7 and TSC controls.	T11_alert_550_modified.json	High	Mappings come from the ruleset; they are not a compliance assessment.
F24	FIM	The alert shows the file owner (uname_after Administrators, uid_after S-1-5-32-544) but not the process or user that made the change.	T11_alert_550_modified.json	High	Real-time mode without whodata; whodata was not tested.
F25	Time	Wazuh and the VM use UTC; the Windows host clock is UTC-7 (script prints local 08:58:10 and UTC 15:58:10); the dashboard shows the browser/host time.	T08_fim_run2_script_output.txt; E46	High	
F26	Volume	The overview showed 484 medium and 391 low severity alerts in 24 hours for one endpoint; only 6 were syscheck events from the test.	E41; E47	Medium	The other alerts were not broken down by rule group.
F27	Hygiene	The VM banner reports Ubuntu 20.04 LTS past the end of standard support (31 May 2025) and "Failed to connect to changelogs.ubuntu.com".	E39	High	
F28	Proposal	A frequency-based custom rule and canary files are proposed for ransomware detection (answer 3).	Report section 8	Low	Proposed only; not deployed or tested in this lab.